

Mohammed Arab

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Education

Algonquin College | Ottawa, Ontario

Diploma of Computer Programming | 05/2023

- Dean's honor list for all academic semesters - 3.97 / 4.0 GPA
- Relevant coursework: Advanced JAVA Programming | Web Programming (LAMP / MERN) | Mobile Application Development | Software Design and Testing

University of Ottawa | Ottawa, Ontario

Bachelors of Health Sciences | 04/2021

- Graduated summa cum laude - 9.31 / 10.0 GPA

Experience

Warner Bros. Discovery - CNN | Ottawa

Software Developer II | 01/2026 - Present

- Contributed to the development of the CNN TV app, using typescript with Lightning.js for webview-based platforms (Samsung, LG, AndroidTV, FireTV, Comcast), brightscript roku, and Swift for TVOS.
- Designed a dual-bundle build system serving an ES6 bundle to modern TV hardware and an ES5 fallback to older devices, cutting bundle TV size by 50% and time-to-first-frame by 60%.
- Maintained Jenkins builds as CI/CD pipelines to ensure a deterministic and consistent release timeline for all different platforms.
- Served as engineering point of contact for design and QA on search, led technical spikes that set implementation direction, and mentored 2 engineers.
- As lead on voice-to-text implementation, I maintained a consistent stream of communication with external contacts (Comcast, Amazon) to ensure feature parity across different platforms.
- Root-caused high-severity production defects on TV hardware with no standard debugging tooling.
- Used agentic tools for AI-accelerated coding with Cursor and Claude Code. This helped ensure all deadlines are met while maintaining high code standards.

Government of Canada - Health Canada | Ottawa, Ontario

IT Developer | 09/2023 - 12/2025

- Developed and maintained the newly released [National Vaccine Catalog \(NVC\)](#) application using Go + Python + SQLite, providing the Canadian public with up-to-date vaccine information.
- Architected the cloud solution for the deploying of the NVC application using Azure web app for containers and storage account services.
- Created the CI/CD pipelines using Azure Devops for NVC to ensure clean and repeatable deployments across environments.
- Communicate daily with stakeholders and business owners to create deployment timelines and explain technical concepts to senior management.
- Ensured proper testing for the NVC application using go tests and Cypress end to end testing framework, allowing the team to catch bugs early on before they reach production.
- Trained, supervised and led a team of 4 developers across timezones working using the Agile scrum methodology.
- Revamped NVC application API to be fully FHIR compatible, allowing EMR systems across the country to connect and share data using a single standard, advancing healthcare interoperability in Canada.

Government of Canada - Canadian Heritage | Ottawa, Ontario

Web Application JAVA Developer | 05/2022 - 09/2023

- Developed and maintained JAVA based web applications open to the Canadian public using JSF + Spring Boot and MSSQL server for database layer.
- Implemented new functionality using multithreading to securely automate the disposal of user data as per information management requirements, saving company time by reducing manual work.
- Worked on improving accessibility by upholding WCAG2.1 AA standards using Web Experience Toolkit (WET) to enable more individuals to use our services.

- Created well tested and well documented applications through the use of JUnit testing framework and writing technical requirements/implementation reports, flow diagrams, and testing documentation.
- Helped maintain [Nomenclature.info](#) by implementing linked open data principles to provide the Nomenclature standard to Museums all over Canada.

Projects

[Thaqalayn.net](#) Hadith Library - Next.js, Deno, GraphQL, NGINX, PostgreSQL, Redis

- Maintain [Thaqalayn.net](#) web application using Next.js for frontend development and Deno as backend JavaScript runtime environment with Typescript to provide services to the frontend. Services include: GraphQL + REST endpoints, Typesense search engine for optimized search results, and NGINX as reverse proxy for load balancing. Postgres is used as an RDBMS. Docker compose is used for deployment.
- Implemented query caching for GQL using Redis to improve web application performance and reduce costs associated with server resource usage.
- Set up synchronization service to automatically update Typesense indexes as application data changes in real-time, providing most up-to-date search results.
- Created backup strategy comprised of weekly automatic backups stored on AWS S3, allowing for database restoration in case of emergencies.
- Experience with Vercel for frontend deployment and AWS EC2 for backend service deployment with Docker. AWS S3 is also used for storage.

[ThaqalaynAPI](#) - React, Node.js, Express.js, MongoDB, Python, Golang

- Created a RESTful API which offered the information found on Thaqalayn.net in JSON format for other programmers.
- Used Node.js + Express.js to create the server, mongoDB to store the data and AWS Lambda was initially used to deploy it. The application is now deployed on a VPS using Caddy server.
- The information was initially scraped off the website using Python. In V2 of the application this was replaced by a Go program for more efficiency.
- Google Gemini AI is used for categorization of ingredients, allowing for the API to provide a list of ingredients and their Islamic statuses in standard JSON format.
- MCP server is available to help AI integration into the API.
- Created a simple [React front-end](#) (deployed on Firebase) to showcase one of the endpoints.
- Received 645,000 requests in February 2023.

[Note Harbor](#) - React, Node.js, Express.js, MongoDB

- Note Harbor is an application that allows for simpler question-based research by allowing users to create notes with specific tags and sources and allows users to filter based on these for easier review in the future.
- [Backend](#) is composed of Node.js + Express server and MongoDB for DB (users also have option of offline use). This service is deployed on a VPS using Caddy server.
- [Frontend](#) composed of a React application developed using Vite and Mantine UI as component library. Deployed on [Netlify](#).
- Work in progress!

[Memory Lane](#) - Golang

- Memory lane is an application that allows partners to create private rooms and post notes that only become available for viewing after some specified period of time, allowing couples to remember and cherish memories together.
- Created as a traditional web application with Golang. The HTML is rendered on the server side before being sent to the client. Other tools used include Go Templ for HTML templating, tailwind CSS for styling, and SQLite for DB.
- Deployed on a VPS using Caddy server.

Languages

English, French, Arabic, CCE French language profile

Skills

JAVA, Javascript, Typescript, Python, SQL, HTML, CSS, React, Node.js, Express.js, Spring / Spring Boot, C#, ASP.NET, JUnit, MySQL, MongoDB, Git/SVN, Maven, Jenkins, Microsoft Azure, LAMP stack, Typesense, NGINX, GraphQL, AWS,

